

Distribution, Determinants, and Normal Reference Values of Thoracic and Abdominal Aortic Diameters by Computed Tomography (from the Framingham Heart Study)

Ian S. Rogers, MD, MPH^{a,b,c}, Joseph M. Massaro, PhD^{d,e}, Quynh A. Truong, MD, MPH^{a,b}, Amir A. Mahabadi, MD^{a,f}, Matthias F. Krieger, BS^a, Caroline S. Fox, MD, MPH^{e,g}, George Thanassoulis, MD, MSc^{e,h,i}, Eric M. Isselbacher, MD^b, Udo Hoffmann, MD, MPH^a, and Christopher J. O'Donnell, MD, MPH^{b,e,*}

Current screening and detection of asymptomatic aortic aneurysms is based largely on uniform cut-point diameters. The aims of this study were to define normal aortic diameters in asymptomatic men and women in a community-based cohort and to determine the association between aortic diameters and traditional risk factors for cardiovascular disease. Measurements of the diameters of the ascending thoracic aorta (AA), descending thoracic aorta (DTA), infrarenal abdominal aorta (IRA), and lower abdominal aorta (LAA) were acquired from 3,431 Framingham Heart Study (FHS) participants. Mean diameters were stratified by gender, age, and body surface area. Univariate associations with risk factor levels were examined, and multivariate linear regression analysis was used to assess the significance of covariate-adjusted relations with aortic diameters. For men, the average diameters were 34.1 mm for the AA, 25.8 mm for the DTA, 19.3 mm for the IRA, and 18.7 mm for the LAA. For women, the average diameters were 31.9 mm for the AA, 23.1 mm for the DTA, 16.7 mm for the IRA, and 16.0 mm for the LAA. The mean aortic diameters were strongly correlated ($p < 0.0001$) with age and body surface area in age-adjusted analyses, and these relations remained significant in multivariate regression analyses. Positive associations of diastolic blood pressure with AA and DTA diameters in both genders and pack-years of cigarette smoking with DTA diameter in women and IRA diameter in men and women were observed. In conclusion, average diameters of the thoracic and abdominal aorta by computed tomography are larger in men compared with women, vary significantly with age and body surface area, and are associated with modifiable cardiovascular disease risk factors, including diastolic blood pressure and cigarette smoking. Published by Elsevier Inc. (Am J Cardiol 2013;111:1510–1516)

^aCardiac MR PET CT Program and ^bDivision of Cardiology, Massachusetts General Hospital and Harvard Medical School, Boston, Massachusetts; ^cDivision of Cardiovascular Medicine, Stanford University School of Medicine, Stanford, California; ^dDepartment of Mathematics, Boston University, Boston, Massachusetts; ^eNational Heart, Lung, and Blood Institute, Framingham Heart Study, Framingham, Massachusetts; ^fDepartment of Cardiology, West-German Heart Center, University Duisburg-Essen, Essen, Germany; ^gDivision of Endocrinology, Brigham and Women's Hospital and Harvard Medical School, Boston, Massachusetts; and Divisions of ^hCardiology and ⁱClinical Epidemiology, McGill University Health Centre, Montreal, Quebec, Canada. Manuscript received October 23, 2012; revised manuscript received and accepted January 20, 2013.

From the Framingham Heart Study (FHS) of the National Heart, Lung, and Blood Institute of the National Institutes of Health, Bethesda, Maryland, and Boston University School of Medicine, Boston, Massachusetts. This work was supported by the National Heart, Lung, and Blood Institute's FHS (Contract N01-HC-25195). Drs. Rogers and Truong received salary support from Grant 1T32 HL076136 from the National Institutes of Health. Dr. Rogers also received salary support from Grant 12CRP12060237 from the American Heart Association. Dr. Truong also received salary support from Grants K23 HL098370 and L30 HL093806 from the National Institutes of Health.

See page 1516 for disclosure information.

*Corresponding author: Tel: 508-935-3435; fax: 508-626-1262.

E-mail address: odonnellc@nhlbi.nih.gov (C.J. O'Donnell).

In individuals without diagnosed aneurysms, there is a paucity of community-based data in large numbers of men and women regarding aortic diameter thresholds for the detection of aortic dilatation above which patients should be monitored or referred for surgical consultation. According to the 2010 guidelines for the diagnosis and management of patients with thoracic aortic disease, an ascending aortic diameter of 5.5 cm is used for most patients without genetically mediated aortic syndromes,¹ despite the knowledge that important patient variables, such as age, gender, and body surface area (BSA), may be associated with normal aortic dimensions and the significance of relative dilatation. Although data are sparse regarding normal diameters of various aortic cross sections in the general population, such data are needed for evaluating the existing standards for aortic aneurysms and for prevention. A recent analysis of the International Registry of Acute Aortic Dissection (IRAD) database found that 60% of patients with dissections had ascending aortic diameters less than the 5.5-cm cut-off value, demonstrating that a “one-size-fits-all” value may not be sufficient to prevent most thoracic aortic dissections.² In this study, we establish the distribution of age- and BSA-specific measurements and cut points for normal aortic diameters assessed by computed tomography

(CT) in a community-based cohort of adult men and women and identify the cardiovascular disease risk factor correlates of large aortic diameters.

Methods

Participants were drawn from an imaging substudy of the community-based Framingham Heart Study (FHS) Offspring Study and Third-Generation Study cohorts. Beginning in 1948, 5,209 men and women aged 28 to 62 years were enrolled in the original cohort of the FHS. The offspring and spouses of the offspring of the original cohort were enrolled in the Offspring Study starting in 1971.^{3,4} Beginning in 2002, 4,095 Third-Generation Study participants, who had ≥ 1 parent in the offspring cohort, were enrolled in the FHS and underwent standard clinic examinations. The standard clinic examination included a physician interview, a physical examination, and laboratory tests, as previously described.⁵

For the present analysis, the substudy sample consisted of Offspring Study and Third-Generation Study participants in the multidetector CT (MDCT) substudy. From June 2002 to April 2005, 3,529 participants underwent MDCT for the assessment of coronary and aortic calcium, 1,418 from the offspring generation and 2,111 from the third generation. Of the 3,529 participants scanned, 3,505 attended offspring exam 7 or generation 3 exam 1. Of these, 3,496 had complete risk factor profiles and were available for analysis. Of the 3,496 participants, 65 (1.9%) had incomplete computed tomographic data. Thus, the final analysis was performed in 3,431 participants of the overall study cohort.

Participant ascertainment for the MDCT study has been previously described.⁶ For inclusion, men were aged ≥ 35 years, women were aged ≥ 40 years and not pregnant, and all participants weighed < 350 lb. The study was approved by the institutional review boards of the Boston University Medical Center and the Massachusetts General Hospital. All participants provided written informed consent.

All participants underwent thoracic electrocardiographically gated, non-contrast-enhanced MDCT in a supine position using an 8-slice multidetector computed tomographic scanner (LightSpeed Ultra; GE Healthcare, Milwaukee, Wisconsin).⁷ In the thorax, contiguous 2.5-mm slices (tube voltage 120 kVp, tube current 320 mA if body weight < 220 lb and 400 mA if body weight > 220 lb, gantry rotation time 500 ms, table feed 3:1) were acquired from the level of the carina to the level of the diaphragm. Two acquisitions were conducted during end-inspiratory breath holds (typical duration 18 seconds). Image acquisition was prospectively triggered in early diastole at 50% of the cardiac cycle. In the abdomen, 25 contiguous 5-mm slices (tube voltage 120 kVp, tube current 400 mA, gantry rotation time 500 ms, table feed 3:1) were acquired covering 125 mm above the level of the first sacral vertebral body.

At each FHS examination, information regarding risk-factor levels was obtained from a history and physical examination obtained by a physician and by laboratory testing, as previously described. Cardiovascular disease outcomes were defined by the presence of coronary heart disease, stroke, peripheral arterial disease, or heart failure and were determined by a 3-physician end point committee, as previously described.⁸ A history of valve surgery or of

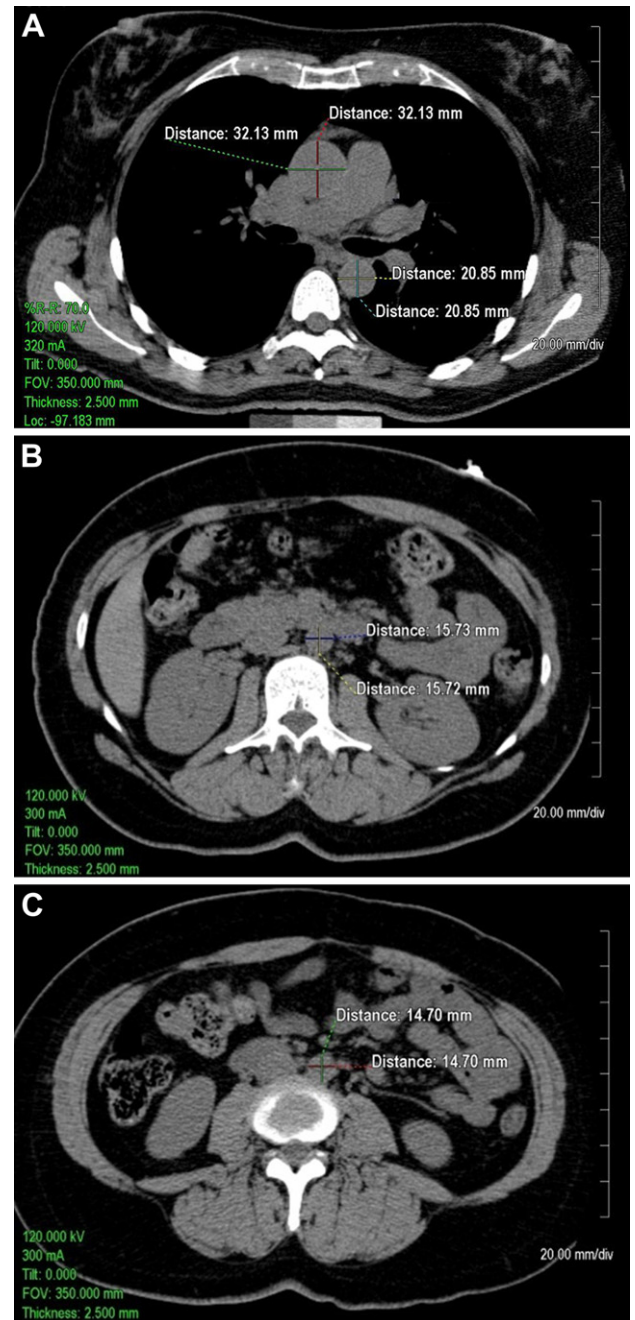


Figure 1. Axial non-contrast-enhanced computed tomographic images (A) of the AA (measurements in upper part of figure) and DTA (measurements in lower part of figure) at the level of the right pulmonary artery, (B) the IRA 5 cm above the aortoiliac bifurcation, and (C) the LAA just above the aortoiliac bifurcation. The images demonstrate sample measurements of the aorta from outside wall to outside wall in the anteroposterior and transverse planes.

surgery of the thoracic aorta or abdominal aorta was obtained by the physician examiner during the interview portion of the participant examination by a physician.

Quantitative measurements of thoracic and abdominal aortic diameters were acquired using the 2.5-mm axial slices acquired during the scan on an Aquarius 3D Workstation (Tera Recon Inc., San Mateo, California). As shown in Figure 1, measurements of the diameters of the ascending

Table 1
Clinical characteristics of the study population

Variable	Overall (n = 3,431)	Women (n = 1,664)	Men (n = 1,767)
Age (yrs)	50.9 ± 10.4	52.2 ± 9.9	49.8 ± 10.7
Body surface area (m ²)	1.9 ± 0.3	1.8 ± 0.2	2.1 ± 0.2
Body mass index >30 kg/m ²	947 (27.6%)	430 (25.8%)	517 (29.3%)
Systolic blood pressure (mm Hg)	122 ± 16	120.3 ± 18	123.6 ± 15
Diastolic blood pressure (mm Hg)	76 ± 9	74 ± 9	78 ± 9
Hypertension	1,006 (29.4%)	443 (26.6%)	563 (31.9%)
Smoking status			
Current	444 (12.9%)	204 (12.3%)	240 (13.6%)
Former	1,345 (39.2%)	704 (42.3%)	641 (36.3%)
Never	1,642 (47.9%)	756 (45.4%)	886 (50.1%)
Smoking amount (pack-years)			
Current*	37.7 ± 21.2	35.5 ± 18.3	40.1 ± 23.9
Former*	21.5 ± 19.7	16.8 ± 16.5	26.4 ± 21.5
Diabetes mellitus	220 (6.4%)	90 (5.4%)	130 (7.4%)
Fasting plasma glucose (mg/dl)	99.2 ± 21.4	95.7 ± 17.9	102.4 ± 23.8
Total cholesterol (mg/dl)	196.4 ± 35.3	198.1 ± 36.3	194.8 ± 34.3
Triglycerides (mg/dl)	128.5 ± 91.5	113.5 ± 68.3	142.6 ± 107.1
High-density lipoprotein cholesterol (mg/dl)	53.3 ± 16.6	61.3 ± 16.8	45.9 ± 12.4
Coronary artery disease	205 (6%)	71 (4.3%)	134 (7.6%)
Valvular heart disease	35 (1%)	14 (0.8%)	21 (1.1%)

Data are expressed as mean ± SD or as number (percentage).

* Data available only for the offspring generation.

thoracic aorta (AA) and descending thoracic aorta (DTA) were acquired on axial slices at the level of the right pulmonary artery. Measurements of the abdominal aorta were acquired on axial slices at 1 slice level 5 cm above the aortoiliac bifurcation (infrarenal abdominal aorta [IRA]) and at 1 slice level above the bifurcation of the abdominal aorta into the common iliac arteries (lower abdominal aorta [LAA]). The measurements were traced manually from outside wall to outside wall of the aorta in the anteroposterior and transverse planes. At each of the 4 locations (i.e., AA, DTA, IRA, and LAA), the mean of anteroposterior and transverse measurements was calculated (mean AA diameter, mean DTA diameter, mean IRA diameter, and mean LAA diameter, respectively). Mean values were used to help control for obliquity of the vasculature on a small number of scans inherent in the use of axial images.

To assess interobserver and intraobserver measurement reliability, reliability measurements were acquired in a random subset of 100 participants (age range 37 to 83 years, 49% women) drawn from the FHS offspring. The random sample included approximately equal numbers of men and women and approximately equal numbers of participants in each of the age groups of 35 to 44, 45 to 54, 55 to 64, 65 to 74, and 75 to 84 years. To evaluate interobserver reliability, 2 of the trained observers performed independent measurements of the 100-participant subset in random order. To evaluate intraobserver reliability, 1 of the observers independently repeated these measurements 1 week later in

random order. The intraobserver intraclass correlation coefficient was >0.97 (range 0.97 to 0.99), suggesting an excellent correlation between reads, with an average difference of 0.5 mm between reads. There was also an excellent interobserver intraclass correlation between readers (all *r* values >0.96, range 0.96 to 0.99), with an average difference of 0.8 mm between readers across the measurements.

Gender-specific, age-adjusted Pearson's correlation coefficients were used to assess univariate correlations between aortic diameters and traditional cardiovascular disease risk factors (age, BSA, systolic blood pressure, diastolic blood pressure, pack-years of smoking, fasting plasma glucose, total cholesterol, triglycerides, and high-density lipoprotein cholesterol). Gender-specific multivariate stepwise linear regression, with age forced into the model, was used to identify traditional risk factors independently associated with aortic diameters; a significance level of 0.05 was used in the stepwise model.

Risk factor data acquired at FHS examinations were then used to create a healthy reference sample after excluding participants with the following conditions from the overall study sample: atherosclerotic cardiovascular disease, hypertension, and previous surgery of the heart valves, thoracic aorta, or abdominal aorta. Mean aortic diameters and percentiles were then calculated for the healthy reference sample, stratified by age (<45, 45 to 54, 55 to 64, and >65 years), gender, and BSA (<1.9, 1.9 to 2.09, and ≥2.1 m² for men and <1.7, 1.7 to 1.89, and ≥1.9 m² for women). Similar calculations were made for the entire sample.

The 90th-percentile aortic diameter was defined at each measurement location. Multivariate logistic regression analysis in the entire sample was used to further identify independent risk factors for dilatation, defined as aortic diameter >90th percentile of the healthy referent cohort. Covariates included in the multivariate model included age, BSA, hypertension, dyslipidemia, smoking, and diabetes. Finally, we used gender- and age-specific simple linear regression of each diameter versus BSA to create nomograms to demonstrate expected aortic diameter versus BSA by gender and age group and to create a formula for calculating the predicted normal range of aortic diameters at each location, accounting for age, gender, and BSA.

Results

As listed in Table 1, the mean age and BSA were 52 ± 10 years and 1.8 ± 0.2 m², respectively, for women and 50 ± 11 years and 2.1 ± 0.2 m², respectively, for men. Approximately 27% of women and 32% of men had hypertension. The healthy referent cohort consisted of a total of 2,343 participants after excluding 1,088 participants with cardiovascular disease, hypertension, and/or previous surgery of heart valves, thoracic aorta, or abdominal aorta.

For all men, the average diameters were 34.1 ± 3.9 mm (range 23.1 to 49.6) for the AA, 25.8 ± 3.0 mm (range 17.9 to 55.2) for the DTA, 19.3 ± 2.9 mm (range 12.8 to 69.9) for IRA, and 18.7 ± 2.7 mm (range 12.9 to 48.5) for the LAA. For all women, the average diameters were 31.9 ± 3.5 mm (range 21.6 to 48.0) for the AA, 23.1 ± 2.6 mm (range 16.8 to 47.2) for the DTA, 16.7 ± 1.8 mm (range

Table 2

Mean and percentiles of the mean diameter of the ascending thoracic aorta (millimeters) by age and body surface area in all women and all men

Age (yrs)	All Women						All Men					
	BSA (m ²)	Mean	25th	50th	75th	90th	BSA (m ²)	Mean	25th	50th	75th	90th
<45	<1.7	28.8	27.0	28.5	30.9	32.1	<1.9	30.3	28.2	30.1	32.5	34.2
	1.7–1.89	30.2	28.0	30.3	31.8	33.8	1.9–2.09	31.1	29.4	31.1	32.5	34.5
	≥1.9	31.1	28.9	31.1	33.2	35.4	≥2.1	33.2	30.8	32.8	34.9	37.0
45–54	<1.7	29.8	27.9	29.7	32.0	33.8	<1.9	32.6	30.1	33.0	34.5	36.4
	1.7–1.89	31.5	29.7	31.5	33.3	35.0	1.9–2.09	33.5	31.1	33.5	35.7	37.4
	≥1.9	32.8	30.8	32.5	35.1	36.8	≥2.1	34.9	32.6	34.5	36.8	39.3
55–64	<1.7	32.3	30.1	32.3	34.0	37.2	<1.9	34.5	32.4	34.7	36.5	38.1
	1.7–1.89	33.5	31.5	33.5	35.3	37.2	1.9–2.09	34.7	32.5	34.3	36.6	39.0
	≥1.9	33.7	31.6	33.6	35.7	37.6	≥2.1	36.8	34.5	36.4	38.6	40.7
≥65	<1.7	33.9	32.1	33.7	35.7	38.0	<1.9	36.2	33.3	35.3	38.4	42.2
	1.7–1.89	33.9	32.3	33.6	35.9	37.9	1.9–2.09	36.7	34.5	36.3	39.0	40.7
	≥1.9	35.1	33.0	34.6	37.1	39.6	≥2.1	38.6	36.1	38.3	41.3	43.4

Table 3

Mean and percentiles of the mean diameter of the descending thoracic aorta (millimeters) by age and body surface area in all women and all men

Age (yrs)	All Women						All Men					
	BSA (m ²)	Mean	25th	50th	75th	90th	BSA (m ²)	Mean	25th	50th	75th	90th
<45	<1.7	20.5	19.5	20.5	21.4	22.2	<1.9	22.5	21.5	22.2	23.4	25.0
	1.7–1.89	21.5	20.5	21.4	22.6	23.6	1.9–2.09	23.5	22.4	23.2	24.4	25.6
	≥1.9	22.2	21.2	22.1	23.2	24.6	≥2.1	24.7	23.2	24.6	25.9	27.0
45–54	<1.7	21.5	20.2	21.5	22.6	23.8	<1.9	24.3	23.2	24.0	25.5	26.8
	1.7–1.89	22.7	21.5	22.5	23.8	24.8	1.9–2.09	24.9	23.6	24.8	26.2	27.5
	≥1.9	23.6	22.2	23.6	24.8	26.0	≥2.1	26.0	24.6	26.0	27.2	28.5
55–64	<1.7	23.1	21.7	23.1	24.4	25.3	<1.9	25.8	24.1	25.6	27.3	28.5
	1.7–1.89	24.0	22.6	24.0	25.1	26.3	1.9–2.09	26.4	25.0	26.3	27.7	28.8
	≥1.9	24.8	23.9	24.8	26.0	27.3	≥2.1	27.7	26.1	27.9	29.2	30.3
≥65	<1.7	25.0	23.2	24.8	26.1	27.7	<1.9	28.4	26.8	28.2	29.6	30.9
	1.7–1.89	25.4	24.2	25.3	27.0	28.1	1.9–2.09	28.8	27.2	28.2	30.1	31.6
	≥1.9	27.1	25.5	26.5	28.4	30.3	≥2.1	30.3	28.4	29.9	32.0	33.0

12.0 to 33.0) for the IRA, and 16.0 ± 1.7 mm (range 11.1 to 33.5) for the LAA.

Gender-specific, age-adjusted univariate Pearson's correlations between continuous measures of cardiovascular disease risk factors and aortic were calculated. The mean diameters of the AA and DTA were correlated ($p < 0.0001$) with age (0.44 to 0.66), BSA (0.31 to 0.41), systolic blood pressure (0.15 to 0.20), and diastolic blood pressure (0.18 to 0.26) in both genders. The mean diameters of the IRA and LAA were correlated ($p < 0.0001$) with age (0.32 to 0.48) and BSA (0.22 to 0.35) in both genders. Weaker correlations (0.03 to 0.14) were noted for abdominal aortic diameters with blood pressure measurements in both genders, with systolic blood pressure (0.09) and diastolic blood pressure (0.14) associated with IRA mean diameter in women and diastolic blood pressure (0.10) associated with LAA mean diameter in women. There were positive correlations of pack-years of cigarette smoking with IRA (0.34, 0.45) and LAA (0.25, 0.28) diameters in men and women and of fasting plasma glucose with IRA (0.08), LAA (0.06), and DTA (0.09) diameters in women and with AA (0.08) and DTA (0.10) diameters in men. Inverse correlations were noted of high-density lipoprotein cholesterol with IRA (−0.08), LAA (−0.07), and DTA (−0.10) diameters in

men and with AA (−0.05) and DTA (−0.09) diameters in women. Paradoxical inverse correlations of very small magnitude were noted for total cholesterol with IRA (−0.07, −0.05) and LAA (−0.08, −0.07) diameters in men and women and with DTA diameter (−0.08) in men.

Multivariate stepwise linear regression analysis was performed between continuous cardiovascular disease risk factors and aortic diameters. For every 1-year increase in age, the mean diameter of the AA was higher by 0.16 mm in women and 0.2 mm in men, the mean DTA diameter was higher by 0.16 mm in women and 0.19 mm in men, the mean IRA diameter was higher by 0.09 mm in women and 0.13 mm in men, and the mean LAA diameter was higher by 0.07 mm in women and 0.12 mm in men. For every 0.1-m² increase in BSA, the mean AA diameter was higher by 4.14 mm in women and 5.8 mm in men, the mean DTA diameter was higher by 3.61 mm in women and 4.15 mm in men, the mean IRA diameter was higher by 2.38 mm in women and 2.92 mm in men, and the mean LAA diameter was higher by 3.11 mm in women and 2.90 mm in men. Of note, for every 1 mm Hg increase in diastolic blood pressure, the mean AA diameter was higher by 0.06 mm in women and 0.08 mm in men, and the mean DTA diameter was higher by 0.03 mm in women and men. The mean IRA

Table 4

Mean and percentiles of the mean diameter of the infrarenal abdominal aorta (millimeters) by age and body surface area in all women and all men

Age (yrs)	All Women						All Men					
	BSA (m ²)	Mean	25th	50th	75th	90th	BSA (m ²)	Mean	25th	50th	75th	90th
<45	<1.7	15.2	14.3	15.2	16.1	16.6	<1.9	16.9	16.1	16.8	17.8	18.5
	1.7–1.89	16.0	15.0	15.9	16.8	17.7	1.9–2.09	17.6	16.8	17.5	18.5	19.5
	≥1.9	16.3	15.4	16.2	17.2	17.8	≥2.1	18.5	17.6	18.5	19.5	20.2
45–54	<1.7	15.8	15.0	15.7	16.5	17.5	<1.9	18.4	17.4	18.5	19.1	20.7
	1.7–1.89	16.6	15.6	16.5	17.6	18.4	1.9–2.09	18.8	17.8	18.7	19.8	20.9
	≥1.9	17.0	16.1	17.0	18.0	19.0	≥2.1	19.4	18.5	19.3	20.3	21.2
55–64	<1.7	16.4	15.2	16.4	17.5	18.6	<1.9	18.9	17.5	18.8	20.1	21.5
	1.7–1.89	17.2	16.3	17.3	18.1	19.0	1.9–2.09	19.4	18.0	19.0	20.6	21.5
	≥1.9	17.6	16.6	17.6	18.5	19.3	≥2.1	20.7	19.0	20.2	21.7	22.9
≥65	<1.7	17.8	16.2	17.5	18.7	20.7	<1.9	21.0	19.1	20.2	21.7	24.6
	1.7–1.89	17.8	16.8	17.7	18.6	20.0	1.9–2.09	21.8	19.5	20.7	22.6	25.5
	≥1.9	18.7	17.1	18.3	19.5	21.8	≥2.1	22.5	19.8	21.4	23.3	26.0

Table 5

Mean and percentiles of the mean diameter of the lower abdominal aorta (millimeters) by age and body surface area in all women and all men

Age (yrs)	All Women						All Men					
	BSA (m ²)	Mean	25th	50th	75th	90th	BSA (m ²)	Mean	25th	50th	75th	90th
<45	<1.7	14.7	13.9	14.7	15.4	16.4	<1.9	16.7	15.7	16.6	17.4	18.1
	1.7–1.89	15.6	14.7	15.7	16.2	17.1	1.9–2.09	17.2	16.4	17.3	18.1	18.8
	≥1.9	16.0	15.0	15.7	16.8	17.8	≥2.1	18.2	17.1	18.1	19.0	19.8
45–54	<1.7	15.2	14.2	15.0	15.9	16.8	<1.9	17.7	16.4	17.4	18.5	20.2
	1.7–1.89	15.8	14.9	15.7	16.7	17.5	1.9–2.09	18.2	17.1	18.1	19.0	20.2
	≥1.9	16.4	15.4	16.2	17.4	18.1	≥2.1	18.8	17.8	18.7	19.8	20.7
55–64	<1.7	15.6	14.6	15.6	16.4	17.3	<1.9	18.4	17.1	18.2	19.5	20.8
	1.7–1.89	16.1	15.2	16.1	16.9	17.4	1.9–2.09	18.4	17.4	18.3	19.3	20.2
	≥1.9	16.9	15.7	16.9	17.6	18.9	≥2.1	19.8	18.3	19.3	20.9	22.5
≥65	<1.7	16.5	15.2	16.5	17.6	18.4	<1.9	20.3	18.0	19.0	20.5	26.5
	1.7–1.89	16.9	15.7	16.6	17.5	18.7	1.9–2.09	21.0	18.5	19.7	21.2	24.9
	≥1.9	17.2	16.1	17.0	17.9	19.4	≥2.1	21.5	19.3	20.9	22.3	25.3

diameter was higher by 0.3 mm in women and 0.27 mm in men with histories of cigarette smoking.

The distributions of aortic diameters by gender, age, and BSA are listed for all men and all women in Tables 2 to 5 for the AA, DTA, IRA, and LAA, respectively. The distributions of aortic diameters by gender, age, and BSA, along with the prevalence of aortic diameters >90th percentile of the healthy referent group by gender, age, and BSA, are listed for men and women in the healthy referent cohort in Supplemental Tables 1a to 1d for the AA, DTA, IRA, and LAA, respectively. Within each gender, diameter increased with increasing age and BSA. Diameters for participants in the healthy referent group were similar to those of the entire cohort across the spectrum of age and BSA categories and measurement locations. Diameters indexed to BSA by age and gender are presented in Supplemental Tables 2a and 2b.

Supplemental Figure 1 provides nomograms of expected aortic diameter ($\pm 95\%$ confidence interval) versus BSA in men and women by age groups for the AA, DTA, IRA, and LAA. The formula for predicting aortic diameter is $14.8 + 0.16(\text{age}) + -1.04(\text{gender}) + 5.34(\text{BSA})$ for the AA, $8.86 + 0.16(\text{age}) + -1.79(\text{gender}) + 4.25(\text{BSA})$ for the DTA, $8.6 + 0.1(\text{age}) + -2.03(\text{gender}) + 2.64(\text{BSA})$ for the

IRA, and $8.71 + 0.08(\text{age}) + -2.11(\text{gender}) + 2.78(\text{BSA})$ for the LAA. The nomograms demonstrate the higher values of aortic diameter with increasing age, BSA, and male gender.

In multivariate-adjusted logistic regression analysis for >90th-percentile thoracic (Supplemental Table 3a) and abdominal (Supplemental Table 3b) aortic diameters, hypertension was independently associated with the upper decile for the AA and DTA in men (odds ratios 2.01 and 1.86, respectively) and women (odds ratios 1.50 and 1.73, respectively). Current or former smoking was independently associated with the upper decile for IRA diameter in women (odds ratio 1.63). There was no association with hyperlipidemia, defined by high cholesterol or use of lipid-lowering treatment.

Discussion

In this study of MDCT in men and women in a community-based cohort, we provide estimates of the strong correlations of aortic diameters with gender, age, and BSA, and we provide gender-specific reference distributions and cut points of thoracic and abdominal aortic diameters stratified by age

and BSA in asymptomatic men and women in the community. We further present normal values of thoracic and abdominal aortic diameters, stratified by gender, age, and BSA, in all participants in the study and in a healthy reference sample of those free of cardiovascular disease, hypertension, and histories of valvular, thoracic aortic, or abdominal aortic surgery. This is the first study that we are aware of to present such community-based data in the thoracic and abdominal aorta.

The few small studies that have characterized normal thoracic diameters in male and female subjects have been largely confined to pediatric and young adult populations, such as a Finnish study in 2003 of 168 children (mean age 11.1 years),⁹ a study in 1993 of 182 United States college students (mean age 21 years),¹⁰ and a 1989 study of 52 children (mean age 9 ± 5 years) and 135 adults (mean age 43 ± 15 years) scanned to exclude structural heart disease.¹¹ Additionally, many studies have measured only thoracic aortic root dimensions, using ultrasound technology.¹² Although ultrasound is reliable in detecting aneurysms in the aortic root and portions of the abdomen, CT and magnetic resonance imaging are frequently used in clinical medicine and allow complete imaging of the entire aorta and therefore may incidentally detect patients with dilated aortas. Moreover, current standard of care for surveillance imaging of aortic dilatation often includes CT or magnetic resonance imaging. As a consequence, physicians must extrapolate normal aortic dimensions in patients with a range of ages, genders, heights, and weights and must often do so using a different modality (i.e., CT or magnetic resonance imaging) than from which the original data were obtained. Comparison of measurements among modalities is confounded by the distinct conventional methods by which measurements are made with each modality.

The FHS MDCT substudy was initially designed with the primary intent of studying the distribution and determinants of coronary calcium and thoracic and abdominal aortic calcium.^{13,14} As such, the available thoracic and abdominal aortic images represented a rich opportunity to examine aortic diameters on MDCT in an unselected community-based cohort. Previous work by Lin et al¹⁵ examined aortic diameter by CT in the thorax alone in 103 subjects who underwent "clinically indicated" computed tomography angiography. A major limitation of that study was the small number of patients. Wolak et al¹⁶ examined aortic diameter by CT in the thorax alone in a cohort that was either self-referred or physician referred or recruited for the Early Identification of Subclinical Atherosclerosis using Noninvasive Imaging Research (EISNER) research protocol. Although potential subjects with known coronary artery disease were excluded, the examined cohort was subject to referral bias. In contrast, the community-based nature of the FHS allowed us to study a truly unselected cohort with reliably collected longitudinal risk factor data that is unbiased by aorta measurements. The span of subject age across the Offspring Study and Third-Generation Study participants provided a robust sample from which generalization to most men and women in the community in whom aortic diameter measurement is feasible. Moreover, we present diameter data for the entire cohort and the healthy referent subset.

We measured and reported AA and DTA diameters at the level of the pulmonary artery, using a commonly used landmark for the measurement of diameters on routine axial images in the thorax. We demonstrated that measurement at this location is highly reproducible, even in the absence of intravenous contrast. Although vascular tortuosity favors 3-dimensional reconstruction to precisely measure areas of interest, particularly in the aortic root, the aorta tends to be the least oblique on axial images at the level of the pulmonary artery. Thus, axial diameters commonly remain the first-line measurements for screening. We chose to measure anteroposterior and transverse diameters in the thorax and the abdomen and report the average diameter to help control for any obliquity of the vessel. However, double oblique measurements should be performed in dilated aortas being monitored for potential surgical repair to control for potential tortuosity and angulation, particularly in the annulus, in the root (sinuses of Valsalva and sinotubular junction), and in the arch, which have an angular orientation in almost all patients. In the clinical setting of known aortic dilatation or disease for which surgical repair may be under consideration, double oblique measurements may have better concordance with planimetry than axial images, particularly in the aortic root.¹⁷ Because the computed tomographic scans in our study were obtained from the level of the carina to the level of the diaphragm, measurements of aortic diameters within the arch were precluded. However, the known prevalence of isolated arch aneurysms is low.

Although aortic diameter was correlated with gender, age, and BSA across all measurements, AA and DTA diameters were also correlated with blood pressure. The strength of correlation with blood pressure did not persist across the abdominal measurements. Instead, the strength of univariate correlation in the abdominal aorta was most notable with number of pack-years in smokers, particularly in current smokers, and with cholesterol values to a lesser extent. These observations fit with theories that thoracic aortic aneurysms, in the absence of select heritable conditions, tend to stem from a hypertensive acceleration of medial degeneration,¹⁸ whereas abdominal aneurysms stem from an atherosclerotic mechanism with tobacco use as an associated exacerbant.¹⁹ However, gender, age, and BSA outweigh the effects of these traditional cardiovascular disease risk factors in predicting aortic diameter. Although we observed paradoxical inverse correlations of increasing cholesterol with aortic diameters, the magnitude of correlation was small, and it is possible that confounding by treatment accounted for these association. Indeed, this hypothesis is supported by the absence of any association of hyperlipidemia, defined by high cholesterol or use of lipid-lowering therapy, with high aortic diameters.

This analysis was limited by the inclusion of only Caucasian participants. Our findings may not generalize to non-Caucasian ethnic and racial groups. Analogous studies in ethnically diverse cohorts, such as Multi-Ethnic Study of Atherosclerosis (MESA), will be needed to provide complementary information. Although the multidetector computed tomographic scans were electrocardiographically gated, allowing much more precise analysis of the AA than with nongated scans, intravenous contrast was not used.

As such, vascular structures were less defined than they would be on contrast-enhanced exams, which may have introduced a small margin of error. Serial imaging of the aorta with CT may result in significant cumulative radiation exposure. According to the 2010 guidelines for the diagnosis and management of patients with thoracic aortic disease, radiation dose should be minimized, and imaging modalities that do not involve exposure to ionizing radiation, such as magnetic resonance imaging, should be considered as alternatives to CT when clinically appropriate.¹

Disclosures

The authors have no conflicts of interest to disclose.

Supplementary Data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.amjcard.2013.01.306>.

- Hiratzka LF, Bakris GL, Beckman JA, Bersin RM, Carr VF, Casey DE Jr, Eagle KA, Hermann LK, Isselbacher EM, Kazerooni EA, Kouchoukos NT, Lytle BW, Milewicz DM, Reich DL, Sen S, Shinn JA, Svensson LG, Williams DM, American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines, American Association for Thoracic Surgery, American College of Radiology, American Stroke Association, Society of Cardiovascular Anesthesiologists, Society for Cardiovascular Angiography and Interventions, Society of Interventional Radiology, Society of Thoracic Surgeons, Society for Vascular Medicine. 2010 ACCF/AHA/AATS/ACR/ASA/SCA/SCAI/SIR/STS/SVM guidelines for the diagnosis and management of patients with thoracic aortic disease. A report of the American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines, American Association for Thoracic Surgery, American College of Radiology, American Stroke Association, Society of Cardiovascular Anesthesiologists, Society for Cardiovascular Angiography and Interventions, Society of Interventional Radiology, Society of Thoracic Surgeons, and Society for Vascular Medicine. *J Am Coll Cardiol* 2010;6:e27–e129.
- Pape LA, Tsai TT, Isselbacher EM, Oh JK, O'gara PT, Evangelista A, Fattori R, Meinhardt G, Trimarchi S, Bossone E, Suzuki T, Cooper JV, Froehlich JB, Nienaber CA, Eagle KA, International Registry of Acute Aortic Dissection (IRAD) Investigators. Aortic diameter $\geq$ 5.5 cm is not a good predictor of type A aortic dissection: observations from the International Registry of Acute Aortic Dissection (IRAD). *Circulation* 2007;116:1120–1127.
- Dawber TR, Kannel WB, Lyell LP. An approach to longitudinal studies in a community: the Framingham Heart Study. *Ann N Y Acad Sci* 1963;107:539–556.
- Kannel WB, Feinleib M, McNamara PM, Garrison RJ, Castelli WP. An investigation of coronary heart disease in families: the Framingham Offspring Study. *Am J Epidemiol* 1979;110:281–290.
- Splansky GL, Corey D, Yang Q, Atwood LD, Cupples LA, Benjamin EJ, D'Agostino RB Sr, Fox CS, Larson MG, Murabito JM, O'Donnell CJ, Vasan RS, Wolf PA, Levy D. The Third Generation Cohort of the National Heart, Lung, and Blood Institute's Framingham Heart Study: design, recruitment, and initial examination. *Am J Epidemiol* 2007;165:1328–1335.
- Fox CS, Massaro JM, Hoffmann U, Pou KM, Maurovich-Horvat P, Liu CY, Vasan RS, Murabito JM, Meigs JB, Cupples LA, D'Agostino RB Sr, O'Donnell CJ. Abdominal visceral and subcutaneous adipose tissue compartments: association with metabolic risk factors in the Framingham Heart Study. *Circulation* 2007;116:39–48.
- Maurovich-Horvat P, Massaro J, Fox CS, Moselewski F, O'Donnell CJ, Hoffmann U. Comparison of anthropometric, area- and volume-based assessment of abdominal subcutaneous and visceral adipose tissue volumes using multi-detector computed tomography. *Int J Obes (Lond)* 2007;31:500–506.
- Murabito JM, Nam BH, D'Agostino RB Sr, Lloyd-Jones DM, O'Donnell CJ, Wilson PW. Accuracy of offspring reports of parental cardiovascular disease history: the Framingham Offspring Study. *Ann Intern Med* 2004;140:434–440.
- Poutanen T, Tikanoja T, Sairanen H, Jokinen E. Normal aortic dimensions and flow in 168 children and young adults. *Clin Physiol Funct Imaging* 2003;23:224–229.
- Reed CM, Richey PA, Pulliam DA, Somes GW, Alpert BS. Aortic dimensions in tall men and women. *Am J Cardiol* 1993;71:608–610.
- Roman MJ, Devereux RB, Kramer-Fox R, O'Loughlin J. Two-dimensional echocardiographic aortic root dimensions in normal children and adults. *Am J Cardiol* 1989;64:507–512.
- Vasan RS, Larson MG, Benjamin EJ, Levy D. Echocardiographic reference values for aortic root size: the Framingham Heart Study. *J Am Soc Echocardiogr* 1995;8:793–800.
- Hoffmann U, Massaro JM, Fox CS, Manders E, O'Donnell CJ. Defining normal distributions of coronary artery calcium in women and men (from the Framingham Heart Study). *Am J Cardiol* 2008;102:1136–1141.
- Preis SR, Hwang SJ, Fox CS, Massaro JM, Levy D, Hoffmann U, O'Donnell CJ. Eligibility of individuals with subclinical coronary artery calcium and intermediate coronary heart disease risk for reclassification (from the Framingham Heart Study). *Am J Cardiol* 2009;103:1710–1715.
- Lin FY, Devereux RB, Roman MJ, Meng J, Jow VM, Jacobs A, Weinsaft JW, Shaw LJ, Berman DS, Gilmore A, Callister TQ, Min JK. Assessment of the thoracic aorta by multidetector computed tomography: age- and sex-specific reference values in adults without evident cardiovascular disease. *J Cardiovasc Comput Tomogr* 2008;2:298–308.
- Wolak A, Gransar H, Thomson LE, Friedman JD, Hachamovitch R, Gutstein A, Shaw LJ, Polk D, Wong ND, Saouaf R, Hayes SW, Rozanski A, Slomka PJ, Germano G, Berman DS. Aortic size assessment by noncontrast cardiac computed tomography: normal limits by age, gender, and body surface area. *JACC Cardiovasc Imaging* 2008;1:200–209.
- Mendoza DD, Kocher M, Devereux RB, Basson CT, Min JK, Holmes K, Dietz HC, Milewicz DM, LeMaire SA, Pyeritz RE, Bavaria JE, Maslen CL, Song H, Kroner BL, Eagle KA, Weinsaft JW; GenTAC (National Registry of Genetically Triggered Thoracic Aortic Aneurysms and Cardiovascular Conditions) Study Investigators. Impact of image analysis methodology on diagnostic and surgical classification of patients with thoracic aortic aneurysms. *Ann Thorac Surg* 2011;92:904–912.
- Carlson RG, Lillehei CW, Edwards JE. Cystic medial necrosis of the ascending aorta in relation to age and hypertension. *Am J Cardiol* 1970;25:411–415.
- Lederle FA, Johnson GR, Wilson SE, Chute EP, Littlooy FN, Bandyk D, Krupski WC, Barone GW, Acher CW, Ballard DJ; Aneurysm Detection and Management (ADAM) Veterans Affairs Cooperative Study Group. Prevalence and associations of abdominal aortic aneurysm detected through screening. *Ann Intern Med* 1997;126:441–449.